

COMPLEX NUMBERS

A general complex number can be written as

$$Z = x + iy$$

where $x, y \in \mathbb{R}$, i.e. are real numbers and $i = \sqrt{-1}$.

- We call x the real part of z and we denote it by $\operatorname{Re}(z)$.
- We call y the imaginary part of z and we denote it by $\operatorname{Im}(z)$.

Complex numbers can be viewed as vectors in the plane

The set of all complex numbers

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\} \text{ is a Vector Space}$$

(of dimension 2) equipped with two operations:

Vector Addition

$$Z_1 + Z_2 = (x_1 + x_2) + i(y_1 + y_2) \text{ for all } Z_1 = x_1 + iy_1, Z_2 = x_2 + iy_2.$$

Scalar Multiplication

$$cZ = cx + i(cy), \text{ for all } c \in \mathbb{R} \text{ and } Z = x + iy.$$

Moreover, complex numbers can do more. Two complex numbers can be multiplied and divided. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

Multiplication

$$z_1 \cdot z_2 = (x_1 + iy_1) \cdot (x_2 + iy_2) = x_1x_2 + ix_1y_2 + iy_1x_2 - y_1y_2 =$$

$$(x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$$

which (unlike vectors) gives another complex number.

Division

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1) \cdot (x_2 - iy_2)}{(x_2 + iy_2) \cdot (x_2 - iy_2)} = \frac{x_1x_2 + y_1y_2 + i(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}$$

$$= \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}, \quad \frac{z_1}{z_2} \in \mathbb{C}$$

Definition The complex number $\bar{z} = x - iy$ is called complex conjugate of $z = x + iy$.

Definition The modulus or magnitude of a complex number

$$z = x + iy \text{ is } |z| = \sqrt{x^2 + y^2}$$

Can be easily verified that:

- $z + \bar{z} = 2 \operatorname{Re}(z)$
- $z - \bar{z} = 2i \operatorname{Im}(z)$

The system of complex numbers, denoted by $\mathbb{C}$ is the set $\mathbb{R}^2$ equipped with addition and complex multiplication.

The notation $z \in \mathbb{C}$ is used to indicate that z belongs to the set of complex numbers.

A field is a set together with two binary operations, the addition and multiplication

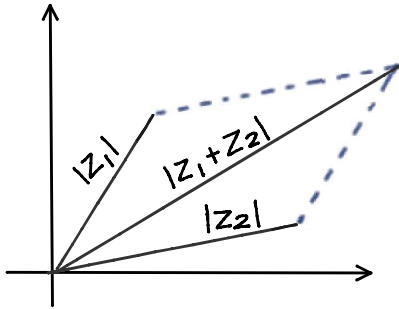
$\mathbb{C}$ is an example of a field

- $z_1 + z_2 = z_2 + z_1$
 - $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$
 - $z + 0 = z$
 - $z + (-z) = 0$
 - $z_1 \cdot z_2 = z_2 \cdot z_1$
 - $(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$
 - $1 \cdot z = z$
 - $z(z^{-1}) = 1, \text{ for } z \neq 0$
 - $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$
- Addition
- multiplication
- (distributivity)

PROPERTIES OF COMPLEX NUMBERS

• Triangle Inequality

For every $z_1, z_2 \in \mathbb{C}$ $|z_1 + z_2| \leq |z_1| + |z_2|$



the magnitude of the sum of two complex numbers is less or equal to the sum of their magnitudes.

- $||z_1| - |z_2|| \leq |z_1 \pm z_2|$

- $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$

- $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad z_2 \neq 0$

Lemma If $z \in \mathbb{C}$ then $|z| = \sqrt{z \cdot \bar{z}}$

Proof: Let $z = x + iy$, then $z \cdot \bar{z} = (x + iy)(x - iy) = x^2 - i^2 y^2 = x^2 + y^2 = |z|^2$.

Lemma If $z_1, z_2 \in \mathbb{C}$ then $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$ and $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$.

Proof: show the above lemma as an exercise.

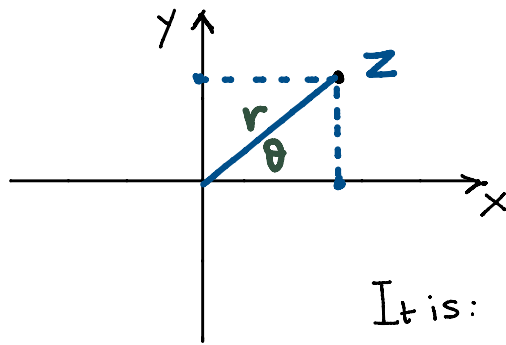
Lemma If $z = \bar{z}$ where z is a complex number then $z \in \mathbb{R}$.

Proof:

Let $z = x + iy$, then $\bar{z} = x - iy$ and $z = \bar{z}$ implies that
 $x + iy = x - iy$ or $2iy = 0$

So $z = x$ is a real.

IN POLAR COORDINATES



The complex plane is also called the Argand diagram.

Let $Z = x + iy \in \mathbb{C}$ and $r = |z|$.

$$\begin{aligned} \text{It is: } x &= \operatorname{Re}(z) = r \cos \theta \\ y &= \operatorname{Im}(z) = r \sin \theta \end{aligned}$$

So, z becomes $Z = r(\cos \theta + i \sin \theta)$

in polar coordinates.

Definition The angle θ is called argument of $z \in \mathbb{C}$ and is denoted by $\arg(z)$.

The argument is given by $\tan \theta = \frac{y}{x}$.

In addition, any complex number is expressed in exponential form:

$$Z = r e^{i\theta}, \text{ where } e^{i\theta} = \cos \theta + i \sin \theta \text{ called } \underline{\text{Euler's formula}}$$

EULER'S IDENTITY

"The most remarkable formula in mathematics"

- Richard Feynman

$$e^{i\pi} + 1 = 0$$

The exponential form is very useful in solving complex algebraic equations, for example:

$$Z^4 = -1 \quad \text{or} \quad Z^3 = \frac{1}{2} - i\frac{3}{2} \quad \text{etc}$$

EXAMPLE

Solve the equation $z^2 = -i$ (1)

We find the exponential form of $-i$:

$$-i = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} = e^{i\frac{3\pi}{2}}, \text{ where } \theta = \frac{3\pi}{2} \text{ when } \theta \in [0, 2\pi).$$

In general:

$$-i = e^{i(\frac{3\pi}{2} + 2\pi m)}, \quad m \in \mathbb{Z}$$

For $m=0$, the equation (1) becomes:

$$z^2 = e^{i\frac{3\pi}{2}} \quad \text{or} \quad z = e^{i\frac{3\pi}{4}}, \text{ which is the first root of (1)}$$

For $m=1$, it is:

$$z^2 = e^{i(\frac{3\pi}{2} + 2\pi)} = e^{i\frac{7\pi}{2}} \quad \text{or} \quad z = e^{i\frac{7\pi}{4}}, \text{ which is the second root of (1).}$$

So, the roots are:

$$Z_1 = e^{i\frac{3\pi}{4}} = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$$

and

$$Z_2 = e^{i\frac{7\pi}{4}} = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2}$$

THE n -th ROOT OF A COMPLEX NUMBER

Proposition [De Moivre]

Let $z = r(\cos \theta + i \sin \theta)$ and n a positive integer. Then,

$$z^n = r^n (\cos n\theta + i \sin n\theta).$$

De Moivre's formula helps in finding the n -th root of any complex number.

The n -th root of unity

Let $z^n = 1$. De Moivre's formula yields:

$$z^n = r^n (\cos n\theta + i \sin n\theta).$$

On the other hand: $1 = 1 + 0 \cdot i = \cos 2k\pi + i \sin 2k\pi$.

Combining the two equations:

$$\begin{cases} r^n \cos n\theta = \cos 2k\pi, & k=0,1,2,\dots \\ r^n \sin n\theta = \sin 2k\pi \end{cases}$$

that by uniqueness in polar representation yields:

$$r=1 \quad \text{and} \quad n\theta = 2k\pi \quad \text{or} \quad \theta = \frac{2k\pi}{n}$$

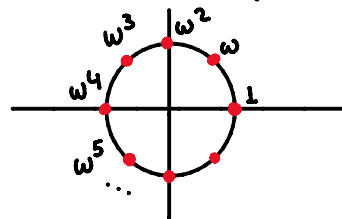
So, $z = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$, $k=0,1,2,\dots,n-1$.

are the n roots of unity.

Remark - link with groups in algebra.

Let $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$. The roots of $z^n = 1$ are:

$$\{1, \omega, \omega^2, \dots, \omega^{n-1}\}$$

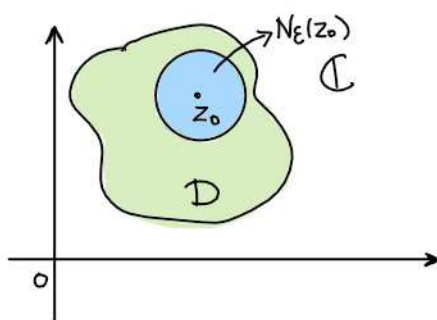


TOPOLOGY OF THE COMPLEX PLANE

Continuity, differentiability and integrability of real one-variable functions required the concepts of open and closed intervals, e.g. (a,b) and $[a,b]$.

Complex functions, however, need open and closed subsets of the plane $\mathbb{C}$.

Let $z_0 = x_0 + iy_0$ be a point in the complex plane. We say that a point z is inside the disk with radius ε centred at z_0 , if $|z - z_0| < \varepsilon$.



$$|z - z_0| < \varepsilon \text{ is equivalent to } (x - x_0)^2 + (y - y_0)^2 < \varepsilon^2$$

All z points inside the circle $|z - z_0| = \varepsilon$

define the neighbourhood of $z_0 \in \mathbb{C}$.

Definition The neighbourhood of $z_0 \in \mathbb{C}$ is the set:

$$N_\varepsilon(z_0) = \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}$$

where $\varepsilon > 0$ is a "small" real positive number.

Definition A subset $D \subset \mathbb{C}$ of the complex plane is called open if every point z_0 in D has a neighbourhood inside D :

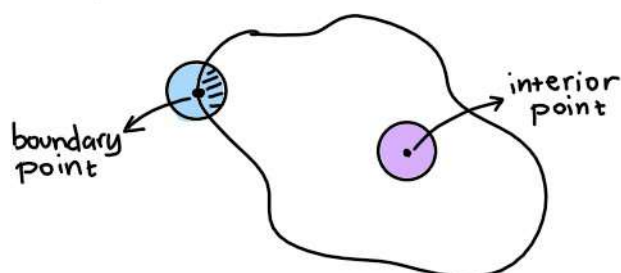
$$\forall z_0 \in D \quad \exists \varepsilon > 0 \text{ such that } N_\varepsilon(z_0) \subset D$$

An open set does not contain "boundary" points, for example $N_\varepsilon(z_0)$ does not contain any of the circle points $|z - z_0| = \varepsilon$.

All points outside of an open set form a "closed" set.

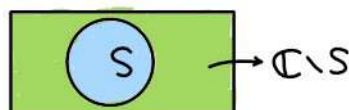
For example, $\mathbb{C} \setminus N_\varepsilon(z_0)$ is a "closed" set and that is due to the circle points $|z - z_0| = \varepsilon$.

- The interior point of a set has a neighbourhood contained entirely within that set.
- Every neighbourhood of a boundary point contains at least one point within the set and one outside it.



- An open set contains only interior points and excludes all boundary points.
- The complement of $S \subset \mathbb{C}$ is the set of all points of $\mathbb{C}$ outside S :

$$\mathbb{C} \setminus S = \{z \in \mathbb{C} : z \notin S\}$$



Definition A subset $S \subset \mathbb{C}$ is a closed set if its complement $\mathbb{C} \setminus S$ is an open set.

Definition A region is a set consisting of an open set and some, none or all of its boundary points.

- A region is bounded if it can be enclosed within a circle.
 is closed if it contains all its boundary points
 and compact if it is closed and bounded.

Notation

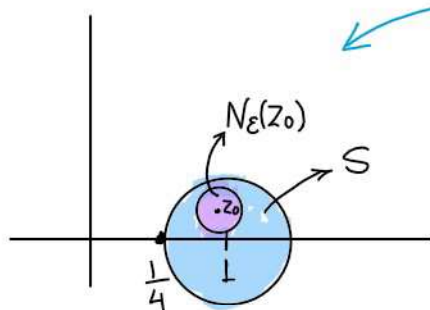
$S = \{z \in \mathbb{C} : |z - a| < r\}$ is an open set

$\bar{S} = \{z \in \mathbb{C} : |z - a| \leq r\}$ is a closed set

$\partial S = \{z \in \mathbb{C} : |z - a| = r\}$ is the boundary of S and of $\bar{S}$.

Also $\bar{S} = S \cup \partial S$

EXAMPLE Let $S = \{z \in \mathbb{C} : |z - 1| < \frac{1}{4}\}$.

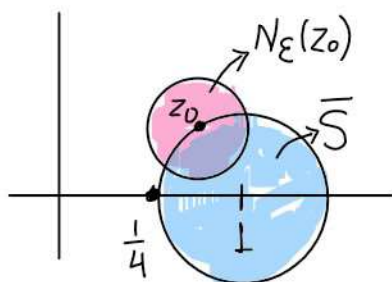


S contains all points inside the circle $|z - 1| = \frac{1}{4}$, that is a circle with radius $\frac{1}{4}$ and centre $(1, 0) = 1 + 0 \cdot i$

- The points of the circle $\partial S = \{z \in \mathbb{C} : |z - 1| = \frac{1}{4}\}$ are the boundary points of the disk S .

- Any point z_0 in S admits a neighbourhood $N_\epsilon(z_0)$ that is entirely inside S . So, S is an open subset of $\mathbb{C}$.

- The set $\bar{S} = \{z \in \mathbb{C} : |z - 1| \leq \frac{1}{4}\}$ is not an open set.



For every z_0 on the circle $|z - 1| = \frac{1}{4}$, there exists a neighbourhood not entirely contained in S and therefore $\bar{S}$ is a closed set.

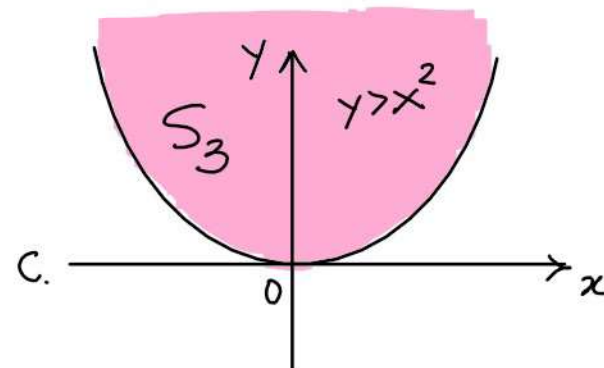
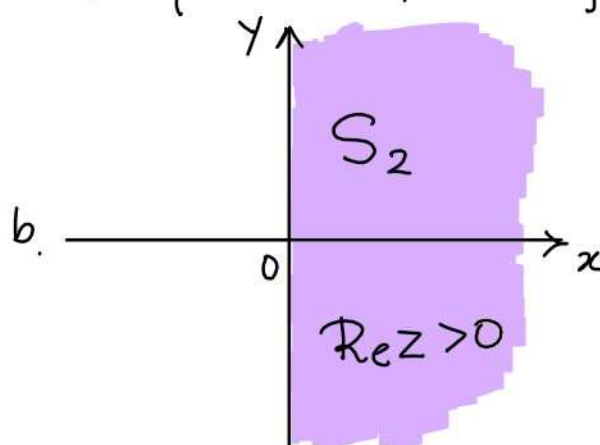
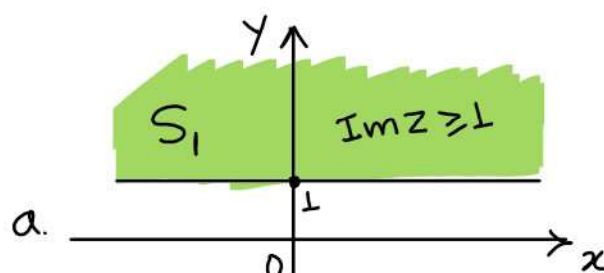
- The boundary ∂S is also a closed set.

FURTHER EXAMPLES

a. $S_1 = \{z \in \mathbb{C} : \operatorname{Im} z \geq 1\}$

b. $S_2 = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$

c. $S_3 = \{z \in \mathbb{C} : y > x^2\}$



a) S_1 is a closed set, since its complement $\mathbb{C} \setminus S_1 = \{z \in \mathbb{C} : \operatorname{Im} z < 1\}$ is an open set.

b) S_2 is an open set. Its boundary is:

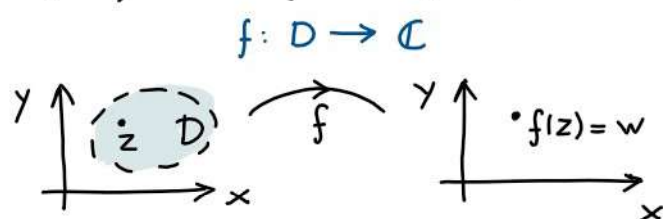
$$\partial S_2 = \{z \in \mathbb{C} : \operatorname{Re} z = 0\} \text{ or } \{z \in \mathbb{C} : x = 0\}.$$

$$\bar{S}_2 = S_2 \cup \partial S_2 = \{z \in \mathbb{C} : \operatorname{Re} z \geq 0\} \text{ is closed.}$$

c) Same for S_3 .

FUNCTION OF A COMPLEX VARIABLE

A function of a complex variable assigns to a complex number z a unique complex number w , namely $f(z) = w$. Let D be an open subset of $\mathbb{C}$, where f is defined



Examples of complex functions

$P(z) = z^3 - 2z^2 + z - 1$	polynomials
$f(z) = \sin z$	trigonometric
$g(z) = e^{kz}$	exponential
$R(z) = \frac{z^2 - z + i}{5z^3 - 2z}$	rational $\frac{P(z)}{Q(z)}$, i.e. the ratio of two polynomials

→ Every complex function $f: D \subset \mathbb{C} \rightarrow \mathbb{C}$ can be expressed in terms of its real and imaginary part.

$$f(z) = f(x+iy) = \underbrace{u(x,y)}_{\text{real}} + i \underbrace{v(x,y)}_{\text{imaginary}}$$

→ f can be considered as a function of two real variables (e.g. x, y).

Examples

1. $f(z) = z^2$ is equivalent to $f(x+iy) = (x+iy)^2 = x^2 + 2ixy - y^2 = (x^2 - y^2) + i(2xy)$

So, $f(x+iy) = u(x,y) + i v(x,y)$, where $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$

2. Let $g(z) = \frac{1}{z+i}$. Then is defined on:
 $D = \{z \in \mathbb{C} : z \neq -i\} = \mathbb{C} \setminus \{-i\}$

and g can be also written as

$$g(z) = \frac{1}{x+i(y+1)} = \frac{x - i(y+1)}{[x+i(y+1)] \cdot [x-i(y+1)]} = \frac{x}{x^2+(y+1)^2} + i \frac{-(y+1)}{x^2+(y+1)^2}$$

real imaginary

3. The exponential of a complex number becomes

$$e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y) = e^x \cos y + i e^x \sin y$$

real imaginary

by applying Euler's formula.

Trigonometric functions of one complex variable.
 It is often convenient to write complex trigonometric functions in their exponential form. From Euler's formula:

$$\left. \begin{array}{l} e^{i\theta} = \cos\theta + i\sin\theta \\ e^{-i\theta} = \cos\theta - i\sin\theta \end{array} \right\} \rightarrow e^{i\theta} + e^{-i\theta} = 2\cos\theta \quad \text{or} \quad \boxed{\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}}$$

where θ is real

↓

and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta \quad \text{or} \quad \boxed{\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}}$

• The complex cosine and sine functions are defined by

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Many trigonometric identities hold for the complex trigonometric functions:

- $\sin^2 z + \cos^2 z = 1$
- $\sin(-z) = -\sin z$, $\cos(-z) = \cos z$
- $\sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2$
- $\cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2$

→ The hyperbolic sin and cos :

$$\sinh z = \frac{e^z - e^{-z}}{2} , \quad \cosh z = \frac{e^z + e^{-z}}{2}$$

Then, • $\cosh^2 z - \sinh^2 z = 1$
and

- $\sin iz = i \sinh z$
- $\cos iz = \cosh z$

Multi-valued functions

Some "functions", like the logarithm and the square root, are defined as the inverse of other functions (e^z , z^2 respectively). It turns out that the inverse of a complex function is not necessarily a single-valued function, but it can be a multi-valued function, which strictly speaking, is not a function.

Multi-valued functions can become single valued when we restrict their domain.

The logarithm function

For any $z \neq 0$, the natural logarithm of z is a complex number w such that $e^w = z$. Let $z = r e^{i\theta}$, then the $\ln z$ can be defined as

$$w = \ln z = \ln(r e^{i\theta}) = \ln r + i(\theta + 2k\pi), \quad k = 0, \pm 1, \pm 2, \dots$$

or as

$$w = \ln z = \ln|z| + i \arg z.$$

where $\ln|z|$ is the usual logarithm of a positive real.

- $\ln z$ (or $\log z$) is a multi-valued function, since for each k

$\ln z = \ln r + i(\theta + 2k\pi)$ has a different value.

When we want to assign one single value to $\ln z$, we often consider the principal-value (or principal branch) of $\ln z$, that is

$$\ln z = \ln r + i\theta, \quad \text{where } 0 \leq \theta < 2\pi$$

for $k=0$.

EXAMPLE Find all values of $\ln i$.

According to the definition, $\ln i = \ln|i| + i\arg(i)$.

$$\ln|i| = \ln 1 = 0$$

and in polar coordinates:

$$i = 0 + i \cdot 1 = \cos\left(\frac{\pi}{2} + 2k\pi\right) + i\sin\left(\frac{\pi}{2} + 2k\pi\right), \quad k=0, \pm 1, \pm 2, \dots$$

$$\text{So, } \arg(i) = \frac{\pi}{2} + 2k\pi \quad \text{and} \quad \ln i = i\left(\frac{\pi}{2} + 2k\pi\right), \quad k=0, \pm 1, \pm 2, \dots$$

In addition, the principal-value of $\ln i$ is $i\frac{\pi}{2}$.

EXAMPLE Find all values of z when $e^{iz} = w$ and w satisfies the quadratic equation:

$$w^2 - iw - 1 = 0.$$

From the quadratic equation we find w :

$$w = \frac{i \pm \sqrt{(-i)^2 - 4(-1)}}{2} = \frac{i \pm \sqrt{3}}{2} = \frac{\sqrt{3}}{2} \pm i\frac{1}{2}$$

Knowing w , we can find z : $iz = \ln w = \ln|w| + i\arg(w)$

$$\text{where } |w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\pm \frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1, \quad \text{and}$$

$$\arg(w) = \arg\left(\frac{\sqrt{3}}{2} \pm i\frac{1}{2}\right) = \pm \frac{\pi}{6} + 2k\pi.$$

Hence,

$$iz = \ln 1 + i\left(\pm \frac{\pi}{6} + 2k\pi\right) \quad \text{or} \quad z = \pm \frac{\pi}{6} + 2k\pi, \quad k=0, \pm 1, \pm 2, \dots$$